

Chapter 3: Digital Awareness

Class 12 – Artificial Intelligence (AI) in Daily Life & Emerging Technologies

Learning Objectives

After completing this lesson, students will be able to:

- Understand the meaning of Artificial Intelligence (AI).
 - Identify examples of AI used in everyday life.
 - Explain how AI is used in education, healthcare, shopping, and transportation.
 - Recognize smart devices powered by AI.
 - Learn about robotics and emerging technologies.
 - Understand the benefits and responsible use of AI.
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Introduction

Artificial Intelligence (AI) is one of the most exciting technologies in today's world. It helps computers and machines perform tasks that usually require human intelligence, such as recognizing speech, answering questions, learning from data, and making decisions.

Today, AI is used in smartphones, smart speakers, online learning platforms, hospitals, transportation, banking, and many other fields. Understanding AI helps students become responsible digital citizens and prepares them for future technologies.

What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) is a technology that enables computers and machines to perform tasks that normally require human intelligence.

AI can:

- Learn from information
- Recognize voices and images
- Answer questions
- Make suggestions

- Solve simple problems
 - Automate repetitive tasks
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AI in Daily Life

Many devices and applications we use every day include AI.

Examples

- Smartphones
 - Voice Assistants
 - Smart TVs
 - Smart Watches
 - Online Shopping Apps
 - Navigation Apps
 - Educational Apps
 - Smart Home Devices
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Voice Assistants

A **voice assistant** is an AI-powered application that responds to voice commands.

Examples of Tasks

- Setting alarms
 - Playing music
 - Answering simple questions
 - Checking the weather
 - Opening applications
 - Making reminders
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AI in Smartphones

Modern smartphones use AI to improve user experience.

Uses

- Face Unlock
 - Voice Typing
 - Camera Enhancement
 - Photo Editing
 - Language Translation
 - Predictive Text
 - Smart Suggestions
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Smart Home Devices

Smart home devices use AI to make homes safer and more convenient.

Examples

- Smart Lights
- Smart Doorbells
- Smart Security Cameras
- Smart Air Conditioners
- Smart Speakers

Benefits

- Save energy
 - Improve home security
 - Easy remote control
 - Greater convenience
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AI in Online Shopping

AI helps online shopping websites provide a better experience.

Uses

- Product Recommendations
 - Personalized Shopping Suggestions
 - Voice Search
 - Chat Support
 - Order Tracking
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AI in Education

AI is making learning more interactive and personalized.

Examples

- Personalized learning recommendations
- Online quizzes
- Smart learning apps
- Language learning tools
- Homework assistance
- Progress tracking

Benefits

- Learn at your own pace
 - Instant feedback
 - Interactive learning
 - Better understanding of concepts
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AI in Healthcare

Hospitals use AI to support doctors and improve patient care.

Examples

- Medical image analysis

- Health monitoring
- Appointment scheduling
- Patient record management
- Health reminder systems

Benefits

- Faster diagnosis support
 - Better patient care
 - Improved efficiency
 - Reduced waiting time
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Face Recognition

Face Recognition is an AI technology that identifies a person's face.

Uses

- Unlocking smartphones
 - Security systems
 - Attendance systems
 - Airport security
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Recommendation Systems

Recommendation systems use AI to suggest content based on user interests.

Examples

- Suggested educational videos
 - Book recommendations
 - Music suggestions
 - Movie recommendations
 - Online shopping suggestions
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Robotics (Introduction)

A **robot** is a machine designed to perform tasks automatically or with human guidance.

Examples

- Cleaning Robots
- Industrial Robots
- Hospital Robots
- Educational Robots

Uses

- Manufacturing
 - Healthcare
 - Education
 - Space Exploration
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Machine Learning (Basic Concept)

Machine Learning is a branch of AI that allows computers to improve by learning from data.

Simple Explanation

The more information a computer receives, the better it can recognize patterns and make useful suggestions.

Emerging Technologies

Emerging technologies are new technologies that are changing the way people live and work.

Examples

- Artificial Intelligence (AI)
- Robotics
- Smart Homes
- Virtual Reality (VR)
- Augmented Reality (AR)
- Internet of Things (IoT)

- Self-Driving Vehicles (Introduction)
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Benefits of Artificial Intelligence

AI offers many advantages.

Benefits

- Saves time
 - Increases efficiency
 - Improves learning
 - Assists in healthcare
 - Makes daily tasks easier
 - Provides smart recommendations
 - Supports innovation
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Responsible Use of AI

Students should use AI responsibly.

Good Practices

- Use AI for learning and creativity.
 - Verify important information from trusted sources.
 - Do not misuse AI tools.
 - Respect privacy and personal information.
 - Use AI under the guidance of parents or teachers when required.
 - Think critically instead of depending only on AI.
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Difference Between Humans and AI

Humans	Artificial Intelligence
Think using experience and emotions	Works using data and programming
Can make personal judgments	Follows programmed instructions and learned patterns
Feel emotions	Does not have real emotions
Learn naturally through life	Learns from data and training

Did You Know?

- AI is used in smartphones, hospitals, schools, banks, and transportation.
- Voice assistants understand spoken commands using AI.
- Many educational apps use AI to recommend lessons based on student progress.
- AI helps people complete many everyday tasks faster and more efficiently.

Points to Remember

- Artificial Intelligence helps machines perform intelligent tasks.
- AI is used in education, healthcare, shopping, communication, and transportation.
- Smart devices use AI to improve user experience.
- Robotics and Machine Learning are important AI technologies.
- AI should always be used responsibly and ethically.
- Human thinking and AI work differently, but both can help solve problems.